

Gedächtnisprotokoll: Data Literacy

Hauptklausur, 8 Fragen, 100 Punkte, 90 Minuten

Solution = I am relatively certain, **Idea** = I am not certain

March 3, 2021

1 Study (6 points)

An experiment was described where a number of n participants was selected. A number m of these participants was then given a vaccine against a certain disease whereas the others did not get the vaccine and served as a control group. Later, observations were made about how many people got the disease.

a) (3 points)

What type of study is this? Observational, active or meta?

Solution: active

b) (3 points)

Identify the independent and dependent variables in this experiment.

Solution:

Independent: participant got the vaccine

Dependent: participant got the disease

2 Information Content (10-12? points)

Given two random variables $X \in [x_1, \dots, x_K]$ and $Y \in [y_1, \dots, y_L]$ with a joint distribution $p(X, Y)$.

a) (3 points)

Give the definition of the information content for outcomes x_k, y_l .

Solution:

$$h(X = x_k, Y = y_l) = -\log_2 p(X = x_k, Y = y_l)$$

b) (3-4? points)

Give the definition of the entropy $\mathbb{H}(p(x, y))$.

Solution:

$$\mathbb{H}(p(x, y)) = -\sum_{k=1}^K \sum_{l=1}^L p(X = x_k, Y = y_l) \cdot \log_2 p(X = x_k, Y = y_l)$$

Note:

To me, there was ambiguity in this question because, in the lecture, we only ever defined $\mathbb{H}(p)$ i.e. over the distribution. While it most likely means the same thing as $\mathbb{H}(p(x, y))$ (defined over an actual value), it nevertheless confused me. Watch out with such small details!

c) (4-5? points)

Prove that if X and Y are independent, then $\mathbb{H}(p(x, y)) = \mathbb{H}(p(x)) + \mathbb{H}(p(y))$.

Solution:

$$\begin{aligned} \mathbb{H}(p(x, y)) &= -\sum_{k=1}^K \sum_{l=1}^L p(X = x_k, Y = y_l) \cdot \log_2 p(X = x_k, Y = y_l) \\ &= -\sum_{k=1}^K \sum_{l=1}^L p(X = x_k)p(Y = y_l) \cdot \log_2 [p(X = x_k)p(Y = y_l)] && \text{independence} \\ &= -\sum_{k=1}^K \sum_{l=1}^L p(x_k)p(y_l) \cdot \log_2 p(x_k) - \sum_{k=1}^K \sum_{l=1}^L p(x_k)p(y_l) \cdot \log_2 p(y_l) \\ &= -\sum_{k=1}^K p(X = x_k) \cdot \log_2 p(X = x_k) - \sum_{l=1}^L p(Y = y_l) \cdot \log_2 p(Y = y_l) && \text{marginalization} \\ &= \mathbb{H}(p(x)) + \mathbb{H}(p(y)) \end{aligned}$$

Note:

Same as before, this was unclear to me if read strictly because of the different definition of $\mathbb{H}(p(x, y))$.

3 Estimation (26 points)

Some random variables $X = [x_1, \dots, x_N]$ were given that were drawn i.i.d. according to a centered Gaussian distribution.

$$p(X; \sigma^2) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-x_i^2}{2\sigma^2}\right)$$

a) (8 points)

Find the Maximum Likelihood Estimator $\hat{\sigma}^2$ for σ^2 .

Solution:

Write down the log-likelihood as follows:

$$\begin{aligned} \ell(X; \sigma^2) &= \log \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-x_i^2}{2\sigma^2}\right) \\ &= \sum_{i=1}^N \frac{-x_i^2}{2\sigma^2} \\ &= -\frac{N \log 2\pi\sigma^2}{2} - \frac{1}{2\sigma^2} \sum_{i=1}^N x_i^2 \end{aligned}$$

Now, we have to maximize this expression by setting the derivative to 0.

$$\begin{aligned} \frac{\partial \ell(X; \sigma^2)}{\partial \sigma^2} &= -\frac{N \log 2\pi\sigma^2}{2} - \frac{1}{2\sigma^2} \sum_{i=1}^N x_i^2 \\ &= \frac{1}{2\sigma^2} \left(-N + \frac{1}{\sigma^2} \sum_{i=1}^N x_i^2 \right) \\ &\stackrel{!}{=} 0 \\ \Rightarrow \hat{\sigma}^2 &= \frac{1}{N} \sum_{i=1}^N x_i^2 \end{aligned}$$

b) (12 points)

Define the Fisher information $I_n(\sigma^2)$ and analytically compute a concrete solution for the given distribution. Remember that for this distribution, we have $\mathbb{E}_{p(X; \sigma^2)}[x_i^2] = \sigma^2$.

Solution:

$$\begin{aligned} I_n(\sigma^2) &= \sum_{i=1}^N \text{var}(s(x_i; \sigma^2)) \\ &= \sum_{i=1}^N \text{var}\left(\frac{\partial \log p(X; \sigma^2)}{\partial \sigma^2}\right) \\ &= \dots \end{aligned}$$

c) (6 points)

Show asymptotic normality by specifying the function towards which the estimator converges using the result from b) for $N \rightarrow \infty$.

Idea

If I understood correctly, the MLE converges towards $\mathcal{N}\left(\sigma^2; \sqrt{\frac{1}{I_n(\sigma^2)}}\right)$, or put differently, for $N \rightarrow \infty$ we have

$$\frac{(\hat{\sigma}^2 - \sigma^2)}{\sqrt{\frac{1}{I_n(\sigma^2)}}} \rightarrow \mathcal{N}(0; 1)$$

There, we can plug in the result from b).

4 Statistical Testing (12-14? points)

A description of a country under pandemic conditions was given where lockdown restrictions could only be lifted once an incidence of $r = 35/100.000$ was reached. A certain county of the country had 32 cases with 69.XXX cases. The mayor was now arguing that due to the small sample size a higher variance was to be expected and that the county was actually reaching an incidence lower than r .

A graph was also provided plotting the binomial probability distribution as well as the survival function under $p(m|n, r)$.

a) (6-8? points)

Using the graph, estimate/read off the p value of a test to confirm this hypothesis.

Idea: (if I understand survival functions correctly)

Using the survival function at $m = 32$, we could read off that the p value was somewhere around 0.0185.

b) (6? points)

Given that the country consists of 401 counties. Provide a rough estimate of how many counties are expected to have a p value like this or more extreme.

Idea:

Because a p value of an observation informally describes how many observations like this or more extreme we can expect, we just have to multiply the p value with the number of experiments. E.g. a p value of 0.05 means that every twentieth experiment can be positive without actually being positive.

This means that in this case we can expect $401 \cdot 0.0185 \approx 7$ counties to have a p value like this or more extreme.

5 Principal Component Analysis (12 points)

Given PCA on some sort of car/motor data with 4 variables (described in detail). How will respective principal components and principal values change under the given scenarios?

a) (7 points)

Due to some error, data was collected in a different scale for $[x_n]_3$. Instead of rotation frequency in Hz (typically 1-100), now rotations per day (1-8640000)).

b) (5 points)

Only motors with a constant number $[x_n]_2 = 48$ were used.

6 Fairness (10 points)

The following probability tables define a joint distribution $p(R, A, Y)$ over the random variables $A \in \{a; b\}$ (sensitive attribute), $Y \in \{0; 1\}$ (true label), and $R \in \{0; 1\}$ (classifier score). Does this distribution satisfy the fairness criterion of *independence*? Why?

$p(A = a, R = 1, Y = 1) = 0.21$	$p(A = a, R = 0, Y = 1) = 0.03$
$p(A = b, R = 1, Y = 0) = 0.19$	$p(A = b, R = 0, Y = 0) = 0.03$
$p(A = a, R = 1, Y = 0) = 0.01$	$p(A = a, R = 0, Y = 0) = 0.25$
$p(A = b, R = 1, Y = 1) = 0.03$	$p(A = b, R = 0, Y = 1) = 0.25$

Solution:

For independence to hold, we need to require $R \perp A$, i.e. $p(R = 1 | A = a) = p(R = 1 | A = b) = p(R = 1)$. We can compute these properties as follows:

$$\begin{aligned} p(A = a) &= \sum_{r \in R, y \in Y} p(A = a, R = r, Y = y) = 0.5 \\ p(A = b) &= 1 - p(A = a) = 0.5 \\ p(R = 1 | A = a) &= \frac{\sum_{y \in Y} P(A = a, R = 1, Y = y)}{P(A = a)} = \frac{0.22}{0.5} = 0.44 \\ p(R = 1 | A = b) &= \frac{\sum_{y \in Y} P(A = b, R = 1, Y = y)}{P(A = b)} = \frac{0.22}{0.5} = 0.44 \\ p(R = 1) &= \sum_{a \in A, y \in Y} p(A = a, R = 1, Y = y) = 0.44 \end{aligned}$$

This means that the distribution satisfies independence.

7 ML Deployment (12 points)

Name four high.level stages of the machine learning deployment workflow and, for each one, describe one major challenge you have to face during that stage.

Solution:

1. **Data Collection**

Challenge: Data is often not available centrally and may be stored in terrible formats

2. **Model Learning**

Challenge: Inefficient training and hyperparameter tuning especially for complex models. Expert knowledge is required.

3. **Model Verification**

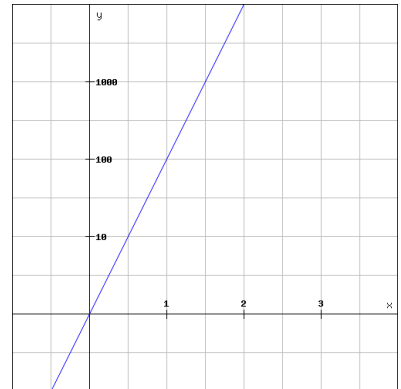
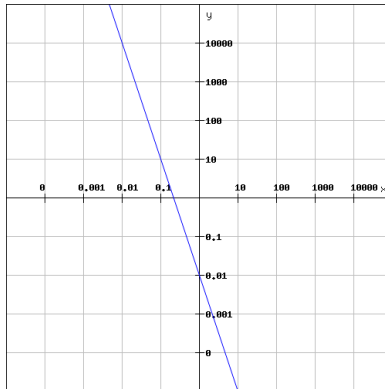
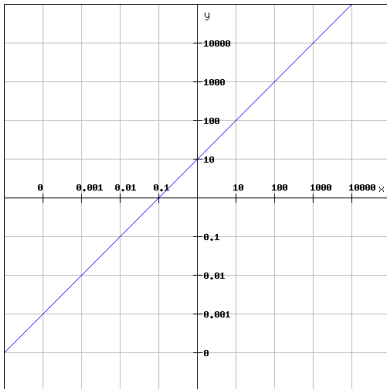
Challenge: formal proofs of correctness are rarely possible and not just code but also data needs to be tested.

4. **Model Deployment**

Challenge: Data set shift can invalidate or deprecate solution and needs to be discovered through monitoring.

8 Plotting (10 points)

Given the following plots, write down the analytical formula for the plotted function. Consider the scale and assume that functions go exactly through the lines.



Solution:

a) $f(x) = 10x$

b) $f(x) = \frac{1}{100x^3}$

c) $f(x) = 10^{2x}$